Public Dissemination Materials Overview

Machine Learning for Seakeeping Control in Waves

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1. Scope and Intended Use

Sungshin Women's University has published the code, data, and documentation from its work on ship-motion forecasting in a public GitHub repository [1]. This overview describes the files available for the research centre's website and explains how to use them.

The two documented models predict six-degree-of-freedom (6-DOF) ship motions: surge, sway, heave, roll, pitch, and yaw. A bidirectional LSTM (BiLSTM) and a causal ModernTCN-style model predict a future motion sequence from an observed history. Both use the same data preparation, hyperparameter-search, and evaluation procedures. The release is intended for researchers and students working on seakeeping, naval engineering, or time-series modelling.

The user manual gives the installation and operating instructions. The theory document explains the model designs and reports earlier experiments. The source files allow users to inspect the implementation and run their own experiments.

2. Available Materials

Material	Contents	Format / location
GitHub repository	Project description, file index, source history, and download links.	GitHub; `README.md` [1]
Final training scripts	`py_BiLSTM_std.py` and `py_modernTCN_std.py`. Each script reads data, constructs and trains a model, searches hyperparameters, evaluates predictions, and saves results.	Python (`.py`) [1]
Code review and user manual	Explanations of functions and tensor shapes; installation, data preparation, training, resumption, inference, troubleshooting, and proposed corrections. English and Korean editions.	Markdown and PDF; English Word file [2]
Theory and earlier experiments	`LRF-SSWU-CodeDocumentation_Theory.pdf`. Forecasting formulation, architecture diagrams, normalization, optimization, search spaces, metrics, and previous results.	PDF, 9 pages [3]
Data and supporting analyses	Motion and wave files in `dataset/` and `dataset_grid/`; the comparison notebook `ComrisonOfModels.ipynb`; and an earlier Causal ModernTCN analysis report.	CSV/OUT, Jupyter notebook, Markdown [1]

This overview is provided as a PDF for publication and a Word file for editing. Section 5 lists the documents to upload and the repository link for access to code and data.

3. Forecasting Methods

Inputs and outputs. The default configuration uses 128 observed samples to predict the next 128 samples of all six motion channels in one forward pass. At the assumed sampling rate of 2 Hz, the forecast covers lead times from 0.5 to 64 seconds after the last observation. Window starts advance by 16 samples. Translations are expressed in metres and rotations in degrees.

Model structures. The BiLSTM processes the observed window in both directions. The final forward and backward hidden states of its last layer are concatenated and passed to a linear forecast head. Both directions read only the observed history. The causal ModernTCN model uses left-padded depthwise temporal convolutions, channel normalization at each time step, pointwise channel mixing, and residual connections. Its temporal projection head uses features from the complete observed window. The theory document describes this project's causal variant; the original LSTM and ModernTCN papers provide the methodological background [3–5].

Training and evaluation. Data are split by file into training, validation, and test sets before windows are generated. Channel means and standard deviations are calculated from training data only. Optuna searches model and training settings using validation MSE, TPE sampling, and median pruning [6]. The default budget is 40 trials, each with up to 40 epochs. The selected configuration is then trained from a fresh initialization for up to 300 epochs. Training and validation remain separate. AdamW updates the weights; validation loss controls learning-rate reduction, early stopping, and checkpoint selection. Gradients are clipped by norm. Test results include MSE and MAE in standardized space and per-DOF RMSE in physical units.

4. Running an Experiment

1. **Obtain the files.** Clone or download the repository. Read the setup instructions in the English manual [2] and the model explanation in the theory PDF [3]. A Korean manual is also available.
2. **Prepare the data.** Supply finite motion values in columns `x1`–`x6`, in the documented order, at 0.5-second intervals. Each training file needs at least 256 rows. Set `ID_DATA_DIR` or place the files in `csv_dataset/`. The published data folders contain different sampling intervals and exported headers; check these and preprocess the data before training. The scripts exclude files beginning with `Wave_`.
3. **Check the environment and run.** Follow the manual's Python/PyTorch setup, data checks, and short synthetic-data example. The scripts select CUDA when available and otherwise use the CPU. Use a separate output path for each experiment.
4. **Keep the results together.** Save the best weights, scaler, split lists, hyperparameters, trial database, training history, metrics, and plots. The manual shows how to reload a saved model for inference. Trained weights are produced by the scripts and are not supplied with this release.

Current limitations. The manual reports short synthetic-data runs and documents two issues that affect experimental use: unequal evaluation batches receive equal weight, and an earlier interruption flag can cause ModernTCN to skip final training on a later run. Corrections are proposed in the manual but have not been applied to the published scripts. The earlier results in the theory PDF were not rerun during documentation preparation. The scripts return point forecasts; prediction intervals are not implemented.

5. Distribution and Citation

The README links to the training scripts, English and Korean manuals, theory PDF, and this overview [1]. Readers can view the Markdown documents on GitHub, download the PDFs, or use the English Word files when preparing edited documents.

The following files are available for publication on the research centre's website:

Item	File or location
Public-materials overview	`LRF_SSWU_Public_Dissemination_Overview.pdf` and the corresponding `.docx`
Theory document	`LRF-SSWU-CodeDocumentation_Theory.pdf`
English user manual	`BiLSTM_ModernTCN_Code_Review_and_User_Manual.md`, with corresponding `.pdf` and `.docx`
Korean user manual	`BiLSTM_ModernTCN_Code_Review_and_User_Manual-kor.md`, with corresponding `.pdf`
Training scripts, data, and notebook	Repository download or individual links in the README [1]

The centre can host the overview, theory document, and manuals and link to GitHub for code, data, and later revisions. The final training scripts reviewed in the manual are identified by commit `31988e994fe6d35e1308760700ecbd2e4f3ec647`. Later documentation changes are recorded in the repository history.

For each experiment, record the code commit, data files, preprocessing, split lists, settings, and software versions. A citation to these materials should identify the institution, document title, repository URL, and the revision or access date. Third-party software has its own attribution and licence information.

6. References

[1] Sungshin Women's University. *Machine Learning for Seakeeping Control in Waves: Public Research Repository.* Code, data, documentation, and project information.
https://github.com/yangjunahn/lrf-sswu

[2] *Detailed Code Review and User Manual for BiLSTM and Causal ModernTCN.* English edition, 19 October 2026. The README also links to the Korean edition and downloadable files.
English manual on GitHub

[3] *Causal ModernTCN and Bi-LSTM for 6-DOF Ship Motion Time Series Prediction and Hyperparameter Optimization: A Code Analysis Report.* Theory and earlier experiments, 9 pages.
Theory PDF on GitHub

[4] Hochreiter, S., and Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Computation*, 9(8), 1735–1780. doi:10.1162/neco.1997.9.8.1735.

[5] Luo, D., and Wang, X. (2024). ModernTCN: A Modern Pure Convolution Structure for General Time Series Analysis. *ICLR 2024*. OpenReview.

[6] Akiba, T., et al. (2019). Optuna: A Next-generation Hyperparameter Optimization Framework. *KDD 2019*. Author preprint, arXiv:1907.10902.

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